# PG No. 13 $C_{4v}$ $4mm$ [ tetragonal ]

Table 1: Harmonics for rank 0.

| No. | polar | axial | expression |
|---|---|---|---|
| 1 | $\mathbb{Q}_0(A_1)$ | $\mathbb{G}_0(A_2)$ | 1 |

Table 2: Harmonics for rank 1.

| No. | polar | axial | expression |
|---|---|---|---|
| 2 | $\mathbb{Q}_1(A_1)$ | $\mathbb{G}_1(A_2)$ | $z$ |
| 3 | $\mathbb{Q}_{1,1}(E)$ | $\mathbb{G}_{1,2}(E)$ | $x$ |
| 4 | $\mathbb{Q}_{1,2}(E)$ | $-\mathbb{G}_{1,1}(E)$ | $y$ |

Table 3: Harmonics for rank 2.

| No. | polar | axial | expression |
|---|---|---|---|
| 5 | $\mathbb{Q}_2(A_1)$ | $\mathbb{G}_2(A_2)$ | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| 6 | $\mathbb{Q}_2(B_1)$ | $\mathbb{G}_2(B_2)$ | $\frac{\sqrt{3}(x-y)(x+y)}{2}$ |
| 7 | $\mathbb{Q}_2(B_2)$ | $\mathbb{G}_2(B_1)$ | $\sqrt{3}xy$ |
| 8 | $\mathbb{Q}_{2,1}(E)$ | $-\mathbb{G}_{2,2}(E)$ | $\sqrt{3}xz$ |
| 9 | $\mathbb{Q}_{2,2}(E)$ | $\mathbb{G}_{2,1}(E)$ | $\sqrt{3}yz$ |

Table 4: Harmonics for rank 3.

| No. | polar | axial | expression |
|---|---|---|---|
| 10 | $\mathbb{Q}_3(A_1)$ | $\mathbb{G}_3(A_2)$ | $-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$ |
| 11 | $\mathbb{Q}_3(B_1)$ | $\mathbb{G}_3(B_2)$ | $\frac{\sqrt{15}z(x-y)(x+y)}{2}$ |
| 12 | $\mathbb{Q}_3(B_2)$ | $\mathbb{G}_3(B_1)$ | $\sqrt{15}xyz$ |
| 13 | $\mathbb{Q}_{3,1}(E,1)$ | $\mathbb{G}_{3,2}(E,1)$ | $\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$ |
| 14 | $\mathbb{Q}_{3,2}(E,1)$ | $-\mathbb{G}_{3,1}(E,1)$ | $-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$ |
| 15 | $\mathbb{Q}_{3,1}(E,2)$ | $\mathbb{G}_{3,2}(E,2)$ | $\frac{\sqrt{15}x(y-z)(y+z)}{2}$ |
| 16 | $\mathbb{Q}_{3,2}(E,2)$ | $-\mathbb{G}_{3,1}(E,2)$ | $\frac{\sqrt{15}y(x-z)(x+z)}{2}$ |

Table 5: Harmonics for rank 4.

| No. | polar | axial | expression |
|---|---|---|---|
| 17 | $\mathbb{Q}_4(A_1,1)$ | $\mathbb{G}_4(A_2,1)$ | $\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$ |
| 18 | $\mathbb{Q}_4(A_1,2)$ | $\mathbb{G}_4(A_2,2)$ | $-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$ |
| 19 | $\mathbb{Q}_4(A_2)$ | $\mathbb{G}_4(A_1)$ | $\frac{\sqrt{35}xy(x-y)(x+y)}{2}$ |
| 20 | $\mathbb{Q}_4(B_1)$ | $\mathbb{G}_4(B_2)$ | $\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$ |
| 21 | $\mathbb{Q}_4(B_2)$ | $\mathbb{G}_4(B_1)$ | $-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$ |
| 22 | $\mathbb{Q}_{4,1}(E,1)$ | $-\mathbb{G}_{4,2}(E,1)$ | $\frac{\sqrt{35}xz(x-z)(x+z)}{2}$ |
| 23 | $\mathbb{Q}_{4,2}(E,1)$ | $\mathbb{G}_{4,1}(E,1)$ | $\frac{\sqrt{35}yz(y-z)(y+z)}{2}$ |
| 24 | $\mathbb{Q}_{4,1}(E,2)$ | $-\mathbb{G}_{4,2}(E,2)$ | $-\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$ |
| 25 | $\mathbb{Q}_{4,2}(E,2)$ | $\mathbb{G}_{4,1}(E,2)$ | $\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$ |

Table 6: Harmonics for rank 5.

| No. | polar | axial | expression |
|---|---|---|---|
| 26 | $\mathbb{Q}_5(A_1,1)$ | $\mathbb{G}_5(A_2,1)$ | $\frac{z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{8}$ |
| 27 | $\mathbb{Q}_5(A_1,2)$ | $\mathbb{G}_5(A_2,2)$ | $\frac{3\sqrt{35}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$ |
| 28 | $\mathbb{Q}_5(A_2)$ | $\mathbb{G}_5(A_1)$ | $\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$ |
| 29 | $\mathbb{Q}_5(B_1)$ | $\mathbb{G}_5(B_2)$ | $-\frac{\sqrt{105}z(x-y)(x+y)\left(x^2+y^2-2z^2\right)}{4}$ |
| 30 | $\mathbb{Q}_5(B_2)$ | $\mathbb{G}_5(B_1)$ | $\frac{\sqrt{105}xyz\left(x^2+y^2-2z^2\right)}{2}$ |
| 31 | $\mathbb{Q}_{5,1}(E,1)$ | $\mathbb{G}_{5,2}(E,1)$ | $\frac{x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{8}$ |
| 32 | $\mathbb{Q}_{5,2}(E,1)$ | $-\mathbb{G}_{5,1}(E,1)$ | $\frac{y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{8}$ |
| 33 | $\mathbb{Q}_{5,1}(E,2)$ | $\mathbb{G}_{5,2}(E,2)$ | $\frac{3\sqrt{35}x\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{8}$ |
| 34 | $\mathbb{Q}_{5,2}(E,2)$ | $-\mathbb{G}_{5,1}(E,2)$ | $\frac{3\sqrt{35}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)}{8}$ |
| 35 | $\mathbb{Q}_{5,1}(E,3)$ | $\mathbb{G}_{5,2}(E,3)$ | $\frac{\sqrt{105}x(y-z)(y+z)\left(2x^2-y^2-z^2\right)}{4}$ |
| 36 | $\mathbb{Q}_{5,2}(E,3)$ | $-\mathbb{G}_{5,1}(E,3)$ | $-\frac{\sqrt{105}y(x-z)(x+z)\left(x^2-2y^2+z^2\right)}{4}$ |

Table 7: Harmonics for rank 6.

| No. | polar | axial | expression |
|---|---|---|---|
| 37 | $\mathbb{Q}_6(A_1,1)$ | $\mathbb{G}_6(A_2,1)$ | $\frac{\sqrt{2}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{8}$ |
| 38 | $\mathbb{Q}_6(A_1,2)$ | $\mathbb{G}_6(A_2,2)$ | $-\frac{\sqrt{14}\left(x^6-15x^4z^2+15x^2z^4+y^6-15y^4z^2+15y^2z^4-2z^6\right)}{8}$ |
| 39 | $\mathbb{Q}_6(A_2)$ | $\mathbb{G}_6(A_1)$ | $-\frac{3\sqrt{7}xy(x-y)(x+y)\left(x^2+y^2-10z^2\right)}{4}$ |
| 40 | $\mathbb{Q}_6(B_1,1)$ | $\mathbb{G}_6(B_2,1)$ | $-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$ |
| 41 | $\mathbb{Q}_6(B_1,2)$ | $\mathbb{G}_6(B_2,2)$ | $\frac{\sqrt{42}(x-y)(x+y)\left(x^4-9x^2y^2-5x^2z^2+y^4-5y^2z^2+5z^4\right)}{8}$ |
| 42 | $\mathbb{Q}_6(B_2,1)$ | $\mathbb{G}_6(B_1,1)$ | $\frac{\sqrt{462}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$ |
| 43 | $\mathbb{Q}_6(B_2,2)$ | $\mathbb{G}_6(B_1,2)$ | $\frac{\sqrt{210}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{16}$ |
| 44 | $\mathbb{Q}_{6,1}(E,1)$ | $-\mathbb{G}_{6,2}(E,1)$ | $-\frac{3\sqrt{7}xz(x-z)(x+z)\left(x^2-10y^2+z^2\right)}{4}$ |
| 45 | $\mathbb{Q}_{6,2}(E,1)$ | $\mathbb{G}_{6,1}(E,1)$ | $\frac{3\sqrt{7}yz(y-z)(y+z)\left(10x^2-y^2-z^2\right)}{4}$ |
| 46 | $\mathbb{Q}_{6,1}(E,2)$ | $-\mathbb{G}_{6,2}(E,2)$ | $\frac{\sqrt{462}xz\left(x^2-3z^2\right)\left(3x^2-z^2\right)}{16}$ |
| 47 | $\mathbb{Q}_{6,2}(E,2)$ | $\mathbb{G}_{6,1}(E,2)$ | $\frac{\sqrt{462}yz\left(y^2-3z^2\right)\left(3y^2-z^2\right)}{16}$ |
| 48 | $\mathbb{Q}_{6,1}(E,3)$ | $-\mathbb{G}_{6,2}(E,3)$ | $\frac{\sqrt{210}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{16}$ |
| 49 | $\mathbb{Q}_{6,2}(E,3)$ | $\mathbb{G}_{6,1}(E,3)$ | $\frac{\sqrt{210}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{16}$ |

Table 8: Harmonics for rank 7.

| No. | polar | axial | expression |
|---|---|---|---|
| 50 | $\mathbb{Q}_7(A_1,1)$ | $\mathbb{G}_7(A_2,1)$ | $-\frac{z\left(35x^6+105x^4y^2-210x^4z^2+105x^2y^4-420x^2y^2z^2+168x^2z^4+35y^6-210y^4z^2+168y^2z^4-16z^6\right)}{16}$ |
| 51 | $\mathbb{Q}_7(A_1,2)$ | $\mathbb{G}_7(A_2,2)$ | $-\frac{\sqrt{231}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)\left(3x^2+3y^2-10z^2\right)}{16}$ |
| 52 | $\mathbb{Q}_7(A_2)$ | $\mathbb{G}_7(A_1)$ | $-\frac{\sqrt{231}xyz(x-y)(x+y)\left(3x^2+3y^2-10z^2\right)}{4}$ |
| 53 | $\mathbb{Q}_7(B_1,1)$ | $\mathbb{G}_7(B_2,1)$ | $\frac{\sqrt{6006}z(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{32}$ |
| 54 | $\mathbb{Q}_7(B_1,2)$ | $\mathbb{G}_7(B_2,2)$ | $\frac{\sqrt{42}z(x-y)(x+y)\left(15x^4+30x^2y^2-80x^2z^2+15y^4-80y^2z^2+48z^4\right)}{32}$ |
| 55 | $\mathbb{Q}_7(B_2,1)$ | $\mathbb{G}_7(B_1,1)$ | $\frac{\sqrt{91}xyz\left(3x^4-5x^2y^2-5x^2z^2+3y^4-5y^2z^2+3z^4\right)}{2}$ |
| 56 | $\mathbb{Q}_7(B_2,2)$ | $\mathbb{G}_7(B_1,2)$ | $-\frac{\sqrt{77}xyz\left(3x^4-20x^2y^2+10x^2z^2+3y^4+10y^2z^2-6z^4\right)}{4}$ |
| 57 | $\mathbb{Q}_{7,1}(E,1)$ | $\mathbb{G}_{7,2}(E,1)$ | $\frac{x\left(16x^6-168x^4y^2-168x^4z^2+210x^2y^4+420x^2y^2z^2+210x^2z^4-35y^6-105y^4z^2-105y^2z^4-35z^6\right)}{16}$ |
| 58 | $\mathbb{Q}_{7,2}(E,1)$ | $-\mathbb{G}_{7,1}(E,1)$ | $-\frac{y\left(35x^6-210x^4y^2+105x^4z^2+168x^2y^4-420x^2y^2z^2+105x^2z^4-16y^6+168y^4z^2-210y^2z^4+35z^6\right)}{16}$ |
| 59 | $\mathbb{Q}_{7,1}(E,2)$ | $\mathbb{G}_{7,2}(E,2)$ | $\frac{\sqrt{231}x\left(10x^2-3y^2-3z^2\right)\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{16}$ |
| 60 | $\mathbb{Q}_{7,2}(E,2)$ | $-\mathbb{G}_{7,1}(E,2)$ | $-\frac{\sqrt{231}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)\left(3x^2-10y^2+3z^2\right)}{16}$ |
| 61 | $\mathbb{Q}_{7,1}(E,3)$ | $\mathbb{G}_{7,2}(E,3)$ | $\frac{\sqrt{6006}x(y-z)(y+z)\left(y^2-4yz+z^2\right)\left(y^2+4yz+z^2\right)}{32}$ |
| 62 | $\mathbb{Q}_{7,2}(E,3)$ | $-\mathbb{G}_{7,1}(E,3)$ | $\frac{\sqrt{6006}y(x-z)(x+z)\left(x^2-4xz+z^2\right)\left(x^2+4xz+z^2\right)}{32}$ |
| 63 | $\mathbb{Q}_{7,1}(E,4)$ | $\mathbb{G}_{7,2}(E,4)$ | $\frac{\sqrt{42}x(y-z)(y+z)\left(48x^4-80x^2y^2-80x^2z^2+15y^4+30y^2z^2+15z^4\right)}{32}$ |
| 64 | $\mathbb{Q}_{7,2}(E,4)$ | $-\mathbb{G}_{7,1}(E,4)$ | $\frac{\sqrt{42}y(x-z)(x+z)\left(15x^4-80x^2y^2+30x^2z^2+48y^4-80y^2z^2+15z^4\right)}{32}$ |

Table 9: Harmonics for rank 8.

| No. | polar | axial | expression |
|---|---|---|---|
| 65 | $\mathbb{Q}_8(A_1,1)$ | $\mathbb{G}_8(A_2,1)$ | $\frac{\sqrt{33}\left(x^8-14x^6y^2-14x^6z^2+35x^4y^4+35x^4z^4-14x^2y^6-14x^2z^6+y^8-14y^6z^2+35y^4z^4-14y^2z^6+z^8\right)}{8}$ |
| 66 | $\mathbb{Q}_8(A_1,2)$ | $\mathbb{G}_8(A_2,2)$ | $\frac{\sqrt{286}\left(x^8-14x^6y^2-14x^6z^2+210x^4y^2z^2-14x^2y^6+210x^2y^4z^2-420x^2y^2z^4+28x^2z^6+y^8-14y^6z^2+28y^2z^6-2z^8\right)}{64}$ |
| 67 | $\mathbb{Q}_8(A_1,3)$ | $\mathbb{G}_8(A_2,3)$ | $\frac{\sqrt{210}\left(x^8-62x^6y^2+34x^6z^2+160x^4y^4-30x^4y^2z^2-80x^4z^4-62x^2y^6-30x^2y^4z^2+60x^2y^2z^4+28x^2z^6+y^8+34y^6z^2-80y^4z^4+28y^2z^6-2z^8\right)}{64}$ |
| 68 | $\mathbb{Q}_8(A_2,1)$ | $\mathbb{G}_8(A_1,1)$ | $\frac{3\sqrt{715}xy(x-y)(x+y)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{16}$ |
| 69 | $\mathbb{Q}_8(A_2,2)$ | $\mathbb{G}_8(A_1,2)$ | $\frac{3\sqrt{77}xy(x-y)(x+y)\left(x^4+2x^2y^2-24x^2z^2+y^4-24y^2z^2+40z^4\right)}{16}$ |
| 70 | $\mathbb{Q}_8(B_1,1)$ | $\mathbb{G}_8(B_2,1)$ | $-\frac{\sqrt{858}(x-y)(x+y)\left(x^2+y^2-14z^2\right)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{64}$ |
| 71 | $\mathbb{Q}_8(B_1,2)$ | $\mathbb{G}_8(B_2,2)$ | $-\frac{3\sqrt{70}(x-y)(x+y)\left(x^6+3x^4y^2-30x^4z^2+3x^2y^4-60x^2y^2z^2+80x^2z^4+y^6-30y^4z^2+80y^2z^4-32z^6\right)}{64}$ |
| 72 | $\mathbb{Q}_8(B_2,1)$ | $\mathbb{G}_8(B_1,1)$ | $-\frac{\sqrt{858}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)\left(x^2+y^2-14z^2\right)}{32}$ |
| 73 | $\mathbb{Q}_8(B_2,2)$ | $\mathbb{G}_8(B_1,2)$ | $-\frac{3\sqrt{70}xy\left(x^6+3x^4y^2-30x^4z^2+3x^2y^4-60x^2y^2z^2+80x^2z^4+y^6-30y^4z^2+80y^2z^4-32z^6\right)}{32}$ |
| 74 | $\mathbb{Q}_{8,1}(E,1)$ | $-\mathbb{G}_{8,2}(E,1)$ | $\frac{3\sqrt{715}xz(x-z)(x+z)\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)}{16}$ |
| 75 | $\mathbb{Q}_{8,2}(E,1)$ | $\mathbb{G}_{8,1}(E,1)$ | $\frac{3\sqrt{715}yz(y-z)(y+z)\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{16}$ |
| 76 | $\mathbb{Q}_{8,1}(E,2)$ | $-\mathbb{G}_{8,2}(E,2)$ | $\frac{3\sqrt{77}xz(x-z)(x+z)\left(x^4-24x^2y^2+2x^2z^2+40y^4-24y^2z^2+z^4\right)}{16}$ |
| 77 | $\mathbb{Q}_{8,2}(E,2)$ | $\mathbb{G}_{8,1}(E,2)$ | $\frac{3\sqrt{77}yz(y-z)(y+z)\left(40x^4-24x^2y^2-24x^2z^2+y^4+2y^2z^2+z^4\right)}{16}$ |
| 78 | $\mathbb{Q}_{8,1}(E,3)$ | $-\mathbb{G}_{8,2}(E,3)$ | $-\frac{\sqrt{858}xz\left(x^2-3z^2\right)\left(3x^2-z^2\right)\left(x^2-14y^2+z^2\right)}{32}$ |
| 79 | $\mathbb{Q}_{8,2}(E,3)$ | $\mathbb{G}_{8,1}(E,3)$ | $\frac{\sqrt{858}yz\left(y^2-3z^2\right)\left(3y^2-z^2\right)\left(14x^2-y^2-z^2\right)}{32}$ |
| 80 | $\mathbb{Q}_{8,1}(E,4)$ | $-\mathbb{G}_{8,2}(E,4)$ | $-\frac{3\sqrt{70}xz\left(x^6-30x^4y^2+3x^4z^2+80x^2y^4-60x^2y^2z^2+3x^2z^4-32y^6+80y^4z^2-30y^2z^4+z^6\right)}{32}$ |
| 81 | $\mathbb{Q}_{8,2}(E,4)$ | $\mathbb{G}_{8,1}(E,4)$ | $\frac{3\sqrt{70}yz\left(32x^6-80x^4y^2-80x^4z^2+30x^2y^4+60x^2y^2z^2+30x^2z^4-y^6-3y^4z^2-3y^2z^4-z^6\right)}{32}$ |

Table 10: Harmonics for rank 9.

| No. | polar | axial | expression |
|---|---|---|---|
| 82 | $\mathbb{Q}_9(A_1,1)$ | $\mathbb{G}_9(A_2,1)$ | $\frac{z\left(315x^8+1260x^6y^2-3360x^6z^2+1890x^4y^4-10080x^4y^2z^2+6048x^4z^4+1260x^2y^6-10080x^2y^4z^2+12096x^2y^2z^4-2304x^2z^6+315y^8-3360y^6z^2+6048y^4z^4-2304y^2z^6+128z^8\right)}{128}$ |
| 83 | $\mathbb{Q}_9(A_1,2)$ | $\mathbb{G}_9(A_2,2)$ | $\frac{3\sqrt{12155}z\left(x^4-4x^3y-6x^2y^2+4xy^3+y^4\right)\left(x^4+4x^3y-6x^2y^2-4xy^3+y^4\right)}{128}$ |
| 84 | $\mathbb{Q}_9(A_1,3)$ | $\mathbb{G}_9(A_2,3)$ | $\frac{3\sqrt{5005}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)\left(x^4+2x^2y^2-8x^2z^2+y^4-8y^2z^2+8z^4\right)}{64}$ |
| 85 | $\mathbb{Q}_9(A_2,1)$ | $\mathbb{G}_9(A_1,1)$ | $-\frac{\sqrt{510510}xyz(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$ |
| 86 | $\mathbb{Q}_9(A_2,2)$ | $\mathbb{G}_9(A_1,2)$ | $\frac{\sqrt{4290}xyz(x-y)(x+y)\left(3x^4-11x^2y^2-7x^2z^2+3y^4-7y^2z^2+7z^4\right)}{8}$ |
| 87 | $\mathbb{Q}_9(B_1,1)$ | $\mathbb{G}_9(B_2,1)$ | $-\frac{\sqrt{4290}z(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)\left(3x^2+3y^2-14z^2\right)}{64}$ |
| 88 | $\mathbb{Q}_9(B_1,2)$ | $\mathbb{G}_9(B_2,2)$ | $-\frac{3\sqrt{110}z(x-y)(x+y)\left(7x^6+21x^4y^2-70x^4z^2+21x^2y^4-140x^2y^2z^2+112x^2z^4+7y^6-70y^4z^2+112y^2z^4-32z^6\right)}{64}$ |
| 89 | $\mathbb{Q}_9(B_2,1)$ | $\mathbb{G}_9(B_1,1)$ | $\frac{\sqrt{330}xyz\left(6x^6-21x^4y^2-21x^4z^2-21x^2y^4+140x^2y^2z^2-21x^2z^4+6y^6-21y^4z^2-21y^2z^4+6z^6\right)}{8}$ |
| 90 | $\mathbb{Q}_9(B_2,2)$ | $\mathbb{G}_9(B_1,2)$ | $\frac{3\sqrt{1430}xyz\left(x^6-7x^4z^2+7x^2z^4+y^6-7y^4z^2+7y^2z^4-2z^6\right)}{8}$ |
| 91 | $\mathbb{Q}_{9,1}(E,1)$ | $\mathbb{G}_{9,2}(E,1)$ | $\frac{x\left(128x^8-2304x^6y^2-2304x^6z^2+6048x^4y^4+12096x^4y^2z^2+6048x^4z^4-3360x^2y^6-10080x^2y^4z^2-10080x^2y^2z^4-3360x^2z^6+315y^8+1260y^6z^2+1890y^4z^4+1260y^2z^6+315z^8\right)}{128}$ |
| 92 | $\mathbb{Q}_{9,2}(E,1)$ | $-\mathbb{G}_{9,1}(E,1)$ | $\frac{y\left(315x^8-3360x^6y^2+1260x^6z^2+6048x^4y^4-10080x^4y^2z^2+1890x^4z^4-2304x^2y^6+12096x^2y^4z^2-10080x^2y^2z^4+1260x^2z^6+128y^8-2304y^6z^2+6048y^4z^4-3360y^2z^6+315z^8\right)}{128}$ |
| 93 | $\mathbb{Q}_{9,1}(E,2)$ | $\mathbb{G}_{9,2}(E,2)$ | $\frac{3\sqrt{12155}x\left(y^4-4y^3z-6y^2z^2+4yz^3+z^4\right)\left(y^4+4y^3z-6y^2z^2-4yz^3+z^4\right)}{128}$ |
| 94 | $\mathbb{Q}_{9,2}(E,2)$ | $-\mathbb{G}_{9,1}(E,2)$ | $\frac{3\sqrt{12155}y\left(x^4-4x^3z-6x^2z^2+4xz^3+z^4\right)\left(x^4+4x^3z-6x^2z^2-4xz^3+z^4\right)}{128}$ |
| 95 | $\mathbb{Q}_{9,1}(E,3)$ | $\mathbb{G}_{9,2}(E,3)$ | $\frac{3\sqrt{5005}x\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)\left(8x^4-8x^2y^2-8x^2z^2+y^4+2y^2z^2+z^4\right)}{64}$ |
| 96 | $\mathbb{Q}_{9,2}(E,3)$ | $-\mathbb{G}_{9,1}(E,3)$ | $\frac{3\sqrt{5005}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)\left(x^4-8x^2y^2+2x^2z^2+8y^4-8y^2z^2+z^4\right)}{64}$ |
| 97 | $\mathbb{Q}_{9,1}(E,4)$ | $\mathbb{G}_{9,2}(E,4)$ | $\frac{\sqrt{4290}x(y-z)(y+z)\left(14x^2-3y^2-3z^2\right)\left(y^2-4yz+z^2\right)\left(y^2+4yz+z^2\right)}{64}$ |
| 98 | $\mathbb{Q}_{9,2}(E,4)$ | $-\mathbb{G}_{9,1}(E,4)$ | $-\frac{\sqrt{4290}y(x-z)(x+z)\left(x^2-4xz+z^2\right)\left(x^2+4xz+z^2\right)\left(3x^2-14y^2+3z^2\right)}{64}$ |
| 99 | $\mathbb{Q}_{9,1}(E,5)$ | $\mathbb{G}_{9,2}(E,5)$ | $\frac{3\sqrt{110}x(y-z)(y+z)\left(32x^6-112x^4y^2-112x^4z^2+70x^2y^4+140x^2y^2z^2+70x^2z^4-7y^6-21y^4z^2-21y^2z^4-7z^6\right)}{64}$ |
| 100 | $\mathbb{Q}_{9,2}(E,5)$ | $-\mathbb{G}_{9,1}(E,5)$ | $-\frac{3\sqrt{110}y(x-z)(x+z)\left(7x^6-70x^4y^2+21x^4z^2+112x^2y^4-140x^2y^2z^2+21x^2z^4-32y^6+112y^4z^2-70y^2z^4+7z^6\right)}{64}$ |

Table 11: Harmonics for rank 10.

| No. | polar | axial | expression |
|---|---|---|---|
| 101 | $\mathbb{Q}_{10}(A_1,1)$ | $\mathbb{G}_{10}(A_2,1)$ | $\frac{\sqrt{390}\left(2x^{10}-45x^8y^2-45x^8z^2+42x^6y^4+1008x^6y^2z^2+42x^6z^4+42x^4y^6-1260x^4y^4z^2-1260x^4y^2z^4+42x^4z^6-45x^2y^8+1008x^2y^6z^2-1260x^2y^4z^4+1008x^2y^2z^6-45x^2z^8+2y^{10}-45y^8z^2+42y^6z^4+42y^4z^6-45y^2z^8+2z^{10}\right)}{96}$ |
| 102 | $\mathbb{Q}_{10}(A_1,2)$ | $\mathbb{G}_{10}(A_2,2)$ | $-\frac{\sqrt{420189}\left(22x^{10}+45x^8y^2-1035x^8z^2-105x^6y^4-630x^6y^2z^2+4935x^6z^4-105x^4y^6+3150x^4y^4z^2-1575x^4y^2z^4-4830x^4z^6+45x^2y^8-630x^2y^6z^2-1575x^2y^4z^4+1260x^2y^2z^6+990x^2z^8+22y^{10}-1035y^8z^2+4935y^6z^4-4830y^4z^6+990y^2z^8-44z^{10}\right)}{35952}$ |
| 103 | $\mathbb{Q}_{10}(A_1,3)$ | $\mathbb{G}_{10}(A_2,3)$ | $-\frac{\sqrt{3213210}\left(3x^{10}-96x^8y^2-39x^8z^2+224x^6y^4+1344x^6y^2z^2-42x^6z^4+224x^4y^6-6720x^4y^4z^2+3360x^4y^2z^4-182x^4z^6-96x^2y^8+1344x^2y^6z^2+3360x^2y^4z^4-2688x^2y^2z^6+135x^2z^8+3y^{10}-39y^8z^2-42y^6z^4-182y^4z^6+135y^2z^8-6z^{10}\right)}{23968}$ |
| 104 | $\mathbb{Q}_{10}(A_2,1)$ | $\mathbb{G}_{10}(A_1,1)$ | $-\frac{\sqrt{12155}xy(x-y)(x+y)\left(x^2+y^2-18z^2\right)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{32}$ |
| 105 | $\mathbb{Q}_{10}(A_2,2)$ | $\mathbb{G}_{10}(A_1,2)$ | $-\frac{\sqrt{2145}xy(x-y)(x+y)\left(x^6+3x^4y^2-42x^4z^2+3x^2y^4-84x^2y^2z^2+168x^2z^4+y^6-42y^4z^2+168y^2z^4-112z^6\right)}{32}$ |
| 106 | $\mathbb{Q}_{10}(B_1,1)$ | $\mathbb{G}_{10}(B_2,1)$ | $-\frac{\sqrt{27170}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)\left(3x^4-11x^2y^2-11x^2z^2+3y^4-11y^2z^2+3z^4\right)}{32}$ |
| 107 | $\mathbb{Q}_{10}(B_1,2)$ | $\mathbb{G}_{10}(B_2,2)$ | $\frac{\sqrt{140063}(x-y)(x+y)\left(22x^8-653x^6y^2-315x^6z^2+2602x^4y^4-945x^4y^2z^2+1575x^4z^4-653x^2y^6-945x^2y^4z^2+3150x^2y^2z^4-1680x^2z^6+22y^8-315y^6z^2+1575y^4z^4-1680y^2z^6+360z^8\right)}{11984}$ |
| 108 | $\mathbb{Q}_{10}(B_1,3)$ | $\mathbb{G}_{10}(B_2,3)$ | $\frac{\sqrt{1071070}(x-y)(x+y)\left(9x^8-165x^6y^2-231x^6z^2-25x^4y^4+3801x^4y^2z^2+406x^4z^4-165x^2y^6+3801x^2y^4z^2-9674x^2y^2z^4+266x^2z^6+9y^8-231y^6z^2+406y^4z^4+266y^2z^6-57z^8\right)}{23968}$ |
| 109 | $\mathbb{Q}_{10}(B_2,1)$ | $\mathbb{G}_{10}(B_1,1)$ | $\frac{\sqrt{92378}xy\left(x^4-10x^2y^2+5y^4\right)\left(5x^4-10x^2y^2+y^4\right)}{256}$ |
| 110 | $\mathbb{Q}_{10}(B_2,2)$ | $\mathbb{G}_{10}(B_1,2)$ | $\frac{\sqrt{4290}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)\left(3x^4+6x^2y^2-96x^2z^2+3y^4-96y^2z^2+224z^4\right)}{256}$ |
| 111 | $\mathbb{Q}_{10}(B_2,3)$ | $\mathbb{G}_{10}(B_1,3)$ | $\frac{\sqrt{165}xy\left(7x^8+28x^6y^2-336x^6z^2+42x^4y^4-1008x^4y^2z^2+1680x^4z^4+28x^2y^6-1008x^2y^4z^2+3360x^2y^2z^4-1792x^2z^6+7y^8-336y^6z^2+1680y^4z^4-1792y^2z^6+384z^8\right)}{128}$ |
| 112 | $\mathbb{Q}_{10,1}(E,1)$ | $-\mathbb{G}_{10,2}(E,1)$ | $-\frac{\sqrt{12155}xz(x-z)(x+z)\left(x^2-18y^2+z^2\right)\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)}{32}$ |
| 113 | $\mathbb{Q}_{10,2}(E,1)$ | $\mathbb{G}_{10,1}(E,1)$ | $\frac{\sqrt{12155}yz(y-z)(y+z)\left(18x^2-y^2-z^2\right)\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{32}$ |
| 114 | $\mathbb{Q}_{10,1}(E,2)$ | $-\mathbb{G}_{10,2}(E,2)$ | $-\frac{\sqrt{2145}xz(x-z)(x+z)\left(x^6-42x^4y^2+3x^4z^2+168x^2y^4-84x^2y^2z^2+3x^2z^4-112y^6+168y^4z^2-42y^2z^4+z^6\right)}{32}$ |
| 115 | $\mathbb{Q}_{10,2}(E,2)$ | $\mathbb{G}_{10,1}(E,2)$ | $\frac{\sqrt{2145}yz(y-z)(y+z)\left(112x^6-168x^4y^2-168x^4z^2+42x^2y^4+84x^2y^2z^2+42x^2z^4-y^6-3y^4z^2-3y^2z^4-z^6\right)}{32}$ |
| 116 | $\mathbb{Q}_{10,1}(E,3)$ | $-\mathbb{G}_{10,2}(E,3)$ | $\frac{\sqrt{92378}xz\left(x^4-10x^2z^2+5z^4\right)\left(5x^4-10x^2z^2+z^4\right)}{256}$ |
| 117 | $\mathbb{Q}_{10,2}(E,3)$ | $\mathbb{G}_{10,1}(E,3)$ | $\frac{\sqrt{92378}yz\left(y^4-10y^2z^2+5z^4\right)\left(5y^4-10y^2z^2+z^4\right)}{256}$ |
| 118 | $\mathbb{Q}_{10,1}(E,4)$ | $-\mathbb{G}_{10,2}(E,4)$ | $\frac{\sqrt{4290}xz\left(x^2-3z^2\right)\left(3x^2-z^2\right)\left(3x^4-96x^2y^2+6x^2z^2+224y^4-96y^2z^2+3z^4\right)}{256}$ |
| 119 | $\mathbb{Q}_{10,2}(E,4)$ | $\mathbb{G}_{10,1}(E,4)$ | $\frac{\sqrt{4290}yz\left(y^2-3z^2\right)\left(3y^2-z^2\right)\left(224x^4-96x^2y^2-96x^2z^2+3y^4+6y^2z^2+3z^4\right)}{256}$ |
| 120 | $\mathbb{Q}_{10,1}(E,5)$ | $-\mathbb{G}_{10,2}(E,5)$ | $\frac{\sqrt{165}xz\left(7x^8-336x^6y^2+28x^6z^2+1680x^4y^4-1008x^4y^2z^2+42x^4z^4-1792x^2y^6+3360x^2y^4z^2-1008x^2y^2z^4+28x^2z^6+384y^8-1792y^6z^2+1680y^4z^4-336y^2z^6+7z^8\right)}{128}$ |
| 121 | $\mathbb{Q}_{10,2}(E,5)$ | $\mathbb{G}_{10,1}(E,5)$ | $\frac{\sqrt{165}yz\left(384x^8-1792x^6y^2-1792x^6z^2+1680x^4y^4+3360x^4y^2z^2+1680x^4z^4-336x^2y^6-1008x^2y^4z^2-1008x^2y^2z^4-336x^2z^6+7y^8+28y^6z^2+42y^4z^4+28y^2z^6+7z^8\right)}{128}$ |

Table 12: Harmonics for rank 11.

| No. | polar | axial | expression |
|---|---|---|---|
| 122 | $\mathbb{Q}_{11}(A_1,1)$ | $\mathbb{G}_{11}(A_2,1)$ | $-\frac{z\left(693x^{10}+3465x^8y^2-11550x^8z^2+6930x^6y^4-46200x^6y^2z^2+36960x^6z^4+6930x^4y^6-69300x^4y^4z^2+110880x^4y^2z^4-31680x^4z^6+3465x^2y^8-46200x^2y^6z^2+110880x^2y^4z^4-63360x^2y^2z^6+7040x^2z^8+693y^{10}-11550y^8z^2+36960y^6z^4-31680y^4z^6+7040y^2z^8-256z^{10}\right)}{256}$ |
| 123 | $\mathbb{Q}_{11}(A_1,2)$ | $\mathbb{G}_{11}(A_2,2)$ | $-\frac{\sqrt{692835}z\left(x^2+y^2-6z^2\right)\left(x^4-4x^3y-6x^2y^2+4xy^3+y^4\right)\left(x^4+4x^3y-6x^2y^2-4xy^3+y^4\right)}{256}$ |
| 124 | $\mathbb{Q}_{11}(A_1,3)$ | $\mathbb{G}_{11}(A_2,3)$ | $-\frac{3\sqrt{1001}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)\left(5x^6+15x^4y^2-70x^4z^2+15x^2y^4-140x^2y^2z^2+168x^2z^4+5y^6-70y^4z^2+168y^2z^4-80z^6\right)}{128}$ |
| 125 | $\mathbb{Q}_{11}(A_2,1)$ | $\mathbb{G}_{11}(A_1,1)$ | $-\frac{\sqrt{692835}xyz(x-y)(x+y)\left(x^2+y^2-6z^2\right)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{32}$ |
| 126 | $\mathbb{Q}_{11}(A_2,2)$ | $\mathbb{G}_{11}(A_1,2)$ | $-\frac{3\sqrt{1001}xyz(x-y)(x+y)\left(5x^6+15x^4y^2-70x^4z^2+15x^2y^4-140x^2y^2z^2+168x^2z^4+5y^6-70y^4z^2+168y^2z^4-80z^6\right)}{32}$ |
| 127 | $\mathbb{Q}_{11}(B_1,1)$ | $\mathbb{G}_{11}(B_2,1)$ | $\frac{\sqrt{1939938}z(x-y)(x+y)\left(x^4-4x^3y-14x^2y^2-4xy^3+y^4\right)\left(x^4+4x^3y-14x^2y^2+4xy^3+y^4\right)}{512}$ |
| 128 | $\mathbb{Q}_{11}(B_1,2)$ | $\mathbb{G}_{11}(B_2,2)$ | $\frac{\sqrt{14586}z(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)\left(15x^4+30x^2y^2-160x^2z^2+15y^4-160y^2z^2+224z^4\right)}{512}$ |
| 129 | $\mathbb{Q}_{11}(B_1,3)$ | $\mathbb{G}_{11}(B_2,3)$ | $\frac{\sqrt{2145}z(x-y)(x+y)\left(21x^8+84x^6y^2-336x^6z^2+126x^4y^4-1008x^4y^2z^2+1008x^4z^4+84x^2y^6-1008x^2y^4z^2+2016x^2y^2z^4-768x^2z^6+21y^8-336y^6z^2+1008y^4z^4-768y^2z^6+128z^8\right)}{256}$ |
| 130 | $\mathbb{Q}_{11}(B_2,1)$ | $\mathbb{G}_{11}(B_1,1)$ | $\frac{\sqrt{2431}xyz\left(5x^8-30x^6y^2-30x^6z^2+63x^4y^4+63x^4z^4-30x^2y^6-30x^2z^6+5y^8-30y^6z^2+63y^4z^4-30y^2z^6+5z^8\right)}{8}$ |
| 131 | $\mathbb{Q}_{11}(B_2,2)$ | $\mathbb{G}_{11}(B_1,2)$ | $-\frac{\sqrt{230945}xyz\left(x^8-6x^6y^2-6x^6z^2+42x^4y^2z^2-6x^2y^6+42x^2y^4z^2-84x^2y^2z^4+12x^2z^6+y^8-6y^6z^2+12y^2z^6-2z^8\right)}{32}$ |
| 132 | $\mathbb{Q}_{11}(B_2,3)$ | $\mathbb{G}_{11}(B_1,3)$ | $-\frac{\sqrt{3003}xyz\left(5x^8-150x^6y^2+90x^6z^2+336x^4y^4-70x^4y^2z^2-168x^4z^4-150x^2y^6-70x^2y^4z^2+140x^2y^2z^4+60x^2z^6+5y^8+90y^6z^2-168y^4z^4+60y^2z^6-10z^8\right)}{32}$ |
| 133 | $\mathbb{Q}_{11,1}(E,1)$ | $\mathbb{G}_{11,2}(E,1)$ | $\frac{x\left(256x^{10}-7040x^8y^2-7040x^8z^2+31680x^6y^4+63360x^6y^2z^2+31680x^6z^4-36960x^4y^6-110880x^4y^4z^2-110880x^4y^2z^4-36960x^4z^6+11550x^2y^8+46200x^2y^6z^2+69300x^2y^4z^4+46200x^2y^2z^6+11550x^2z^8-693y^{10}-3465y^8z^2-6930y^6z^4-6930y^4z^6-3465y^2z^8-693z^{10}\right)}{256}$ |
| 134 | $\mathbb{Q}_{11,2}(E,1)$ | $-\mathbb{G}_{11,1}(E,1)$ | $-\frac{y\left(693x^{10}-11550x^8y^2+3465x^8z^2+36960x^6y^4-46200x^6y^2z^2+6930x^6z^4-31680x^4y^6+110880x^4y^4z^2-69300x^4y^2z^4+6930x^4z^6+7040x^2y^8-63360x^2y^6z^2+110880x^2y^4z^4-46200x^2y^2z^6+3465x^2z^8-256y^{10}+7040y^8z^2-31680y^6z^4+36960y^4z^6-11550y^2z^8+693z^{10}\right)}{256}$ |
| 135 | $\mathbb{Q}_{11,1}(E,2)$ | $\mathbb{G}_{11,2}(E,2)$ | $\frac{\sqrt{692835}x\left(6x^2-y^2-z^2\right)\left(y^4-4y^3z-6y^2z^2+4yz^3+z^4\right)\left(y^4+4y^3z-6y^2z^2-4yz^3+z^4\right)}{256}$ |
| 136 | $\mathbb{Q}_{11,2}(E,2)$ | $-\mathbb{G}_{11,1}(E,2)$ | $-\frac{\sqrt{692835}y\left(x^2-6y^2+z^2\right)\left(x^4-4x^3z-6x^2z^2+4xz^3+z^4\right)\left(x^4+4x^3z-6x^2z^2-4xz^3+z^4\right)}{256}$ |
| 137 | $\mathbb{Q}_{11,1}(E,3)$ | $\mathbb{G}_{11,2}(E,3)$ | $\frac{3\sqrt{1001}x\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)\left(80x^6-168x^4y^2-168x^4z^2+70x^2y^4+140x^2y^2z^2+70x^2z^4-5y^6-15y^4z^2-15y^2z^4-5z^6\right)}{128}$ |
| 138 | $\mathbb{Q}_{11,2}(E,3)$ | $-\mathbb{G}_{11,1}(E,3)$ | $-\frac{3\sqrt{1001}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)\left(5x^6-70x^4y^2+15x^4z^2+168x^2y^4-140x^2y^2z^2+15x^2z^4-80y^6+168y^4z^2-70y^2z^4+5z^6\right)}{128}$ |
| 139 | $\mathbb{Q}_{11,1}(E,4)$ | $\mathbb{G}_{11,2}(E,4)$ | $\frac{\sqrt{1939938}x(y-z)(y+z)\left(y^4-4y^3z-14y^2z^2-4yz^3+z^4\right)\left(y^4+4y^3z-14y^2z^2+4yz^3+z^4\right)}{512}$ |
| 140 | $\mathbb{Q}_{11,2}(E,4)$ | $-\mathbb{G}_{11,1}(E,4)$ | $\frac{\sqrt{1939938}y(x-z)(x+z)\left(x^4-4x^3z-14x^2z^2-4xz^3+z^4\right)\left(x^4+4x^3z-14x^2z^2+4xz^3+z^4\right)}{512}$ |
| 141 | $\mathbb{Q}_{11,1}(E,5)$ | $\mathbb{G}_{11,2}(E,5)$ | $\frac{\sqrt{14586}x(y-z)(y+z)\left(y^2-4yz+z^2\right)\left(y^2+4yz+z^2\right)\left(224x^4-160x^2y^2-160x^2z^2+15y^4+30y^2z^2+15z^4\right)}{512}$ |
| 142 | $\mathbb{Q}_{11,2}(E,5)$ | $-\mathbb{G}_{11,1}(E,5)$ | $\frac{\sqrt{14586}y(x-z)(x+z)\left(x^2-4xz+z^2\right)\left(x^2+4xz+z^2\right)\left(15x^4-160x^2y^2+30x^2z^2+224y^4-160y^2z^2+15z^4\right)}{512}$ |
| 143 | $\mathbb{Q}_{11,1}(E,6)$ | $\mathbb{G}_{11,2}(E,6)$ | $\frac{\sqrt{2145}x(y-z)(y+z)\left(128x^8-768x^6y^2-768x^6z^2+1008x^4y^4+2016x^4y^2z^2+1008x^4z^4-336x^2y^6-1008x^2y^4z^2-1008x^2y^2z^4-336x^2z^6+21y^8+84y^6z^2+126y^4z^4+84y^2z^6+21z^8\right)}{256}$ |
| 144 | $\mathbb{Q}_{11,2}(E,6)$ | $-\mathbb{G}_{11,1}(E,6)$ | $\frac{\sqrt{2145}y(x-z)(x+z)\left(21x^8-336x^6y^2+84x^6z^2+1008x^4y^4-1008x^4y^2z^2+126x^4z^4-768x^2y^6+2016x^2y^4z^2-1008x^2y^2z^4+84x^2z^6+128y^8-768y^6z^2+1008y^4z^4-336y^2z^6+21z^8\right)}{256}$ |